AMAN SANADHYA

amansanadhya@utexas.edu, sanadhyaman25@gmail.com • +1-(737)-296-3674
https://amansanadhya.github.io/

EDUCATION

The University of Texas at Austin Jan 2025 – Present
Master's of Science in Mechanical Engineering
Area of Specialization- Dynamic Systems and Controls

Institute of Technology, Nirma University, Ahmedabad, Gujarat Oct 2020 – Jun 2024
Bachelor's of Technology in Mechanical Engineering.
Minor Specialization- Robotics and Automation Engineering.

EXPERIENCES

The University of Texas at Austin Aug 2025 – Present
Graduate Research Assistant
- Developing automated systems for drilling engineering applications in the oil and gas industry.
- Conducted research on soft GI tract phantoms using dielectric elastomers for capsule endoscopy applications.

Graduate Teaching Assistant
- Teaching Assistant for ME 314D – Dynamics (Fall 2025): Led discussion sections and guided students in analytical and applied problem-solving.

Skillonation Edtech Jun 2024 – Dec 2024
Robotics Researcher and Mentor
- Mentored school students in hands-on STEM robotics projects.
- Advanced AGV designs for hospital logistics and delivery applications.
- Worked on a collaborative project with the Indian Space Research Organization (ISRO) on satellite launching and space exploration.

Indian Institute of Technology, Jodhpur Jan 2024 – May 2024
Research Intern
- Conducted experimental and computational analysis of dielectric elastomer actuators (DEAs) under electro-mechanical loading conditions.
- First author of a research manuscript on planar DEA design for applications in soft robotics.

DCM Shriram Industries Ltd., Kota Jun 2023 – Aug 2023
Mechanical Engineering Intern
- Learned rayon manufacturing processes and preventive maintenance workflows.
- Resolved mechanical issues in looms, ply cablers, and cord-coning machinery.

Shree Balajee Infra Equipment Pvt. Ltd., Jaipur Jun 2022 – Jul 2022
Mechanical Site Trainee
- Observed operation and maintenance of IC engines in heavy machinery.
- Gained practical exposure to hydraulic systems and client interactions.

PUBLICATIONS

- **Sanadhya A.,**& Trivedi R. *"Experimental and Machine Learning–Based Parametric Study of Dielectric Elastomer Bending Actuators for Finger-like gripping"*. physica status solidi (a) (Special Edition) [Accepted for Publication], 2026.
- **Sanadhya A.**, Chandak K., Gohil J., Trivedi R. *"Advancements in finger prosthesis: A design using shape memory alloy actuation mechanism"*, Results in Engineering, Elsevier, 2025. DOI: 10.1016/j.rineng.2025.105050
- Chandak K., **Sanadhya A.,** Gohil J., Trivedi R., Parikh P., *"Soft finger-like actuator actuated using electromyography signals"*. International Journal on Interactive Design and Manufacturing(IJIDeM), Springer, 2024. DOI: 10.1007/s12008-024-01911-1
- **Sanadhya A.,**& Trivedi R. *"Nonlinear Deformation Behavior of Circular Dielectric Elastomer Actuators: A Parametric Study Based on Gent's Elasticity Theory"*. [Submitted], 2025.

CONFERENCES

- **Sanadhya A.,** & Trivedi R., *"Experimental and Machine Learning–Based Parametric Study of Dielectric Elastomer Bending Actuators for Finger-like gripping"*, In Proceedings of Recent Advancements in Materials Science and Nanotechnology (RAMAN), Ahmedabad, India, February 2026
- **Sanadhya A.,** Bigelow E., Tummalacherla B. & Xia F., *"Parametric Evaluation of Hemispherical Balloon Dielectric Elastomer Actuator (DEA) for Gastrointestinal Tract Phantom Motion"*, In Proceedings of International Conference on Advanced Intelligent Mechatronics (AIM), Genova, Italy, July 2026

PROJECTS

- **Automated Drill Mud Mixing System for the Oil & Gas Industry** — Jan 2026 – Present
  Designing and programming a Level 4 automated control system for the drill mud mixing process used in oil and gas well drilling, focusing on process reliability, safety, and efficiency.
- **Development of Soft GI Tract Phantom Using Dielectric Elastomers** — Mar 2025 — Present
  Designed and modeled a bio-inspired actuator using dielectric elastomer (DE) materials to replicate the peristaltic motion and local deformation of the Gastrointestinal (GI) tract for capsule endoscopy research.
- **Modeling and Analysis of Dielectric Elastomer Actuator** — Jan 2024 – May 2024
  Exploration, comparison, and design assessment of Dielectric Elastomer Actuator and analyzing experimental & analytical data using mathematical material modeling for quasi-static electro-mechanical loading.
- **Different Actuation Mechanisms for Upper limb Prosthesis** — Jul 2023 – Jan 2024
  Successful actuation of Shape Memory Alloy and pneumatic based finger-like-actuator using ElectroMyography sensors showing applications in upper limb prosthesis. Helped solving problems like irritation, bulkiness, and discomfort caused by powered prosthesis.
- **Two-wheeled Self Balancing Robot using Arduino and MPU6050** — Apr 2023 – May 2023
  Developed a self-balancing robot with two wheels from scratch, employing gyroscope principles. Utilized MPU 6050 and Arduino Uno to integrate gyroscopic effects with accelerometers.
- **Soft Robotic Pneumatic Assistive glove actuator** — Dec 2022 – Mar 2023
  Developed soft robotics solutions to aid stroke and motor neuron disease patients in daily tasks and neuron connection restoration. Designed an advanced soft robotic glove hand gripper integrating EMG, FSR, and Flex sensors for improved functionality.
- **Solar Panel Tracking System using Simulink** — Nov 2022 –Dec 2022
  Modeled a physical system using simulink and PID controllers after tracing the Path of the Sun for a particular season and made the solar panel follow the motion of the Sun.

RESEARCH INTEREST

- Dynamic Systems, Controls and Automation
- Soft Robotics and Biomechanics applications
- Soft and Smart Materials actuators

SKILLS

- Control Implementation
- Electroactive Material
- Mathematical & Analytical Modeling
- Optimization & Estimation
- Prosthetics & Assistive Devices
- Experimental systems & instrumentation
- Robotics & Mechatronics
- Programming Tools
- Biomechanical devices

ACTIVITIES

- **Joint Secretary**, Mechanical Engineering Students' Association (MESA) Board'23
- **Core Organizing Committee Member**, PRAVEG'22 & PRAVEG'23
- **Writer & Director** Winning Drama team in ANVESHAN' 22, ITNU
- **2× State level Bronze Medalist and 1× District level Gold Medalist** in Taekwondo
- **Runners-up** U-19 Zonal Volleyball tournament
- **Bronze Medalist** in National Science Olympiad and National Computer Olympiad